Problem Set 8 – Multiple Regression Part V

Problem 1 (20 points)

Please access the *Life Expectancy* dataset. These data are from Kaggle. The Global Health Observatory (GHO) data repository under World Health Organization (WHO) keeps track of the health status as well as many other related factors for all countries. Those data are made available to the public for the purpose of health data analysis. These data are from 193 countries and were collected from the same WHO data repository website. The data was collected from WHO and the United Nations website with the help of Deeksha Russell and Duan Wang. The dataset consists of:

Response:

- life expectancy

Regressors:

- adult Mortality
- infant deaths
- alcohol
- percentage expenditure
- hepatitis B
- measles
- BMI
- under-five deaths
- polio
- total expenditure
- diphtheria
- HIV/AIDS
- GDP
- Population

Please answer each of the following questions. So that your answers remain focused, you are to include ONLY the item(s) requested (else, your earned credit will be reduced).

a. (5 points) Regress "life expectancy" on the fourteen regressors listed here. Copy and paste the Summary of Fit table and Parameter Estimates table showing the VIFs in the space below

Term	Estimate	Std Error	t Ratio	Prob>|t|	VIF
Intercept	63.884851	0.643718	99.24	<.0001*	.
Adult Mortality	-0.023028	0.001105	-20.84	<.0001*	1.6875215
infant deaths	0.1230713	0.012679	9.71	<.0001*	206.61394
Alcohol	0.3428187	0.03198	10.72	<.0001*	1.4611879
percentage expenditure	-0.000116	0.000216	-0.54	0.5925	12.716373
Hepatitis B	-0.007429	0.005341	-1.39	0.1645	1.6459446
Measles	-1.023e-5	1.287e-5	-0.80	0.4267	1.4823833
BMI	0.0917944	0.006315	14.54	<.0001*	1.3693283
under-five deaths	-0.09456	0.009199	-10.28	<.0001*	197.63171
Polio	0.0186246	0.006176	3.02	0.0026*	1.6917623
Total expenditure	0.1441016	0.04874	2.96	0.0032*	1.1053671
Diphtheria	0.0330198	0.007087	4.66	<.0001*	2.0580919
HIV/AIDS	-0.457342	0.02143	-21.34	<.0001*	1.470663
GDP	0.0001418	3.367e-5	4.21	<.0001*	13.141328
Population	1.9667e-9	2.103e-9	0.94	0.3498	1.9319073

b. (5 points) Infant Deaths shows up with a very large VIF. Drop that regressor and rerun the model. Copy and paste the Summary of Fit table and Parameter Estimates table showing the new VIFs in the space below.

Term	Estimate	Std Error	t Ratio	Prob>|t|	VIF
Intercept	63.366883	0.659536	96.08	<.0001*	.
Adult Mortality	-0.024605	0.001124	-21.90	<.0001*	1.6510539
Alcohol	0.2922618	0.03244	9.01	<.0001*	1.4224291
percentage expenditure	-0.000154	0.000222	-0.69	0.4894	12.712224
Hepatitis B	-0.008314	0.005491	-1.51	0.1302	1.645465
Measles	2.2528e-5	1.277e-5	1.76	0.0778	1.3804153
BMI	0.0945051	0.006486	14.57	<.0001*	1.36665
under-five deaths	-0.005802	0.001036	-5.60	<.0001*	2.3696255
Polio	0.0221856	0.006338	3.50	0.0005*	1.6857924
Total expenditure	0.1335544	0.050098	2.67	0.0078*	1.1048177
Diphtheria	0.0410035	0.007237	5.67	<.0001*	2.0303686
HIV/AIDS	-0.459265	0.022031	-20.85	<.0001*	1.4705373
GDP	0.0001483	3.461e-5	4.28	<.0001*	13.136162
Population	7.1082e-9	2.092e-9	3.40	0.0007*	1.8093212

RSquare	0.746231
RSquare Adj	0.744213
Root Mean Square Error	4.449032
Mean of Response	69.3023
Observations (or Sum Wgts)	1649

c. (5 points) Refer to parts a. and b. In three sentences or less, explain generally how the numerical value and statistical significance of each of the parameter estimates changed?

After removing Infant Deaths, most parameter estimates changed slightly, while Under-five Deaths changed substantially from -0.09456 to -0.005802. Most variables maintained their statistical significance, although Population became statistically significant ($p = 0.0007$), while Measles remained nonsignificant despite changing from a negative to a positive coefficient. Overall, removing Infant Deaths reduced multicollinearity and changed some coefficient estimates without substantially altering the significance of most predictors.

d. (5 points) Refer to part c. Keep Infant Deaths out of the model and remove GDP as well (largest VIF at this stage). Rerun. In two sentences or less, compare the fit (as measured by R^2) of the full model with that of the model with GDP and Infant Deaths removed. [1]

Removing Infant Deaths and GDP decreased R^2 from 0.7601 to 0.7417, indicating a slight reduction in model fit. However, the reduced model still explains approximately 74.17% of the variation in life expectancy.

Problem 2 (30 points)

[1] In preparing this problem I read a very strange article that appeared in Academic Medicine and Surgery. The authors focused on what an acceptable value of R^2 would be, I guess to referees. Their conclusions were based on a survey of published studies that reported R^2. They concluded, $R^2 > 0.15$. Anything can get published if you find the right niche!

Please access the *Brazilian Rent* dataset which will be linked on Canvas. This dataset contains 10962 houses to rent with 13 different features:

1. City where the property is located.
2. **Property area.**
3. **Number of rooms.**
4. **Number of bathrooms.**
5. **Number of parking spaces.**
6. **Floor where apartment is located.**
7. Accept animals?
8. Furnished?
9. Homeowners' association tax.
10. Rent amount.
11. Property tax.
12. Fire insurance.
13. **Total rental cost.**

Please answer each of the following questions. So that your answers remain focused, you are to include ONLY the item(s) requested (else, your earned credit will be reduced).

a. (2 points) Use JMP to fit a model with variables 2 - 6 as regressors and variable 13 as the response. Generate the hii values for these data and save those to your JMP data table. Order the data table by the hii values, largest to smallest. Copy and paste the Summary of Fit table, the Parameter Estimates table, and a snippet of the ordered data table with the first 10 rows displayed, in the space below.

	city	area	rooms	bathroom	parking spaces	floor	animal	furniture	hoa (R$)	rent amount (R$)	property tax (R$)	fire insurance (R$)	total (R$)	h total (R$)
1	Belo Horizonte	46335	4	8	5	11	accept	furnished	960	8500	646	114	10220	0.7308694066
2	Belo Horizonte	80	3	2	2	301	accept	not furnished	750	2600	164	35	3549	0.2914398728
3	SÃ£o Paulo	24606	5	4	4	12	accept	not furnished	2254	8100	7859	103	18320	0.2060039719
4	Campinas	12732	3	2	0	3	accept	not furnished	700	1600	96	21	2417	0.0559612093
5	SÃ£o Paulo	900	4	9	8	1	accept	not furnished	0	15000	5700	226	20930	0.0068586466
6	Campinas	64	2	2	2	51	accept	not furnished	800	1900	129	25	2854	0.0066334388
7	Belo Horizonte	156	7	2	1	3	accept	not furnished	350	2200	187	30	2767	0.0058100616
8	Belo Horizonte	355	3	4	8	8	accept	not furnished	2700	6000	1106	80	9886	0.0057850455
9	SÃ£o Paulo	43	2	7	2	11	accept	not furnished	257	2270	0	29	2556	0.0055668861
10	SÃ£o Paulo	353	4	5	5	46	accept	not furnished	3000	10000	0	127	13130	0.0055160897

RSquare	0.041058
RSquare Adj	0.040475
Root Mean Square Error	18230.01
Mean of Response	5458.473
Observations (or Sum Wgts)	8231

Term	Estimate	Std Error	t Ratio	Prob>|t|
Intercept	552.08464	549.5626	1.00	0.3151
area	0.0522346	0.339832	0.15	0.8778
rooms	-1124.775	290.7055	-3.87	0.0001*
bathroom	3559.4516	276.9625	12.85	<.0001*
parking spaces	21.641657	249.0334	0.09	0.9308
floor	25.913164	33.39619	0.78	0.4378

b. (3 points) Focus on the four observations with the largest hii values (one or two orders of magnitude bigger than all the rest). In two sentences or less, indicate which of the six regression variables seem to be different for those four.

Area and Floor appear to be the unusual regression variables among the four observations with the highest leverage. Three observations have exceptionally large areas (12,732–46,335), while another has an unusually high floor value of 301.

c. (5 points) Refer to part b. Use the Exclude/Unexclude toggle to set aside those four observations and rerun the model. Copy and paste the Summary of Fit table and the Parameter Estimates table in the space below.

RSquare	0.046338
RSquare Adj	0.045758
Root Mean Square Error	18183.54
Mean of Response	5456.933
Observations (or Sum Wgts)	8227

Term	Estimate	Std Error	t Ratio	Prob>|t|
Intercept	1343.7081	578.1678	2.32	0.0201*
area	23.269917	3.472619	6.70	<.0001*
rooms	-1713.801	304.2147	-5.63	<.0001*
bathroom	2733.2033	302.9411	9.02	<.0001*
parking spaces	-278.2634	252.6433	-1.10	0.2708
floor	32.52702	39.57688	0.82	0.4112

d. (3 points) Refer to part c. In three sentences or less, explain generally what has changed with the fit, and with the numerical value and significance of the parameter estimates.

After removing the four high-leverage observations, R^2 increased slightly from 0.0411 to 0.0463, indicating a small improvement in model fit. The parameter estimates changed, particularly Area, which increased from 0.0522 to 23.2699 and became statistically significant ($p < 0.0001$). Rooms and Bathrooms remained statistically significant, while Parking Spaces and Floor remained nonsignificant.

e. (5 points) Leave the four observations out that you omitted in part c. Generate the studentized residuals, save those to your JMP data table. Order the data table by the studentized residuals, largest to smallest. Copy and paste a snippet of the ordered data table with the first 10 rows displayed, in the space below.

	city	area	rooms	bathroom	parking spaces	floor	animal	furniture	hoa (R$)	rent amount (R$)	property tax (R$)	fire insurance (R$)	total (R$)	h total (R$)	Studentized Resid total (R$)
1	Belo Horizonte	155	1	4	0	4	not accept	not furnished	1117000	2790	64	38	1120000	0.0028682285	60.896767689
2	Belo Horizonte	155	1	4	0	4	not accept	not furnished	1117000	2790	64	38	1120000	0.0028682285	60.896767689
3	SÃ£o Paulo	42	1	2	1	6	not accept	not furnished	690	2500	313700	32	316900	0.0005420682	17.103370766
4	SÃ£o Paulo	340	5	4	2	7	accept	not furnished	220000	12000	1000	153	233200	0.0011985293	12.212644453
5	SÃ£o Paulo	285	4	5	4	6	accept	furnished	200000	20000	1834	254	222100	0.0008763757	11.4567625
6	Rio de Janeiro	35	1	1	0	1	accept	furnished	81150	4500	9900	58	95610	0.0004452648	5.0827431086
7	Porto Alegre	42	1	1	0	10	accept	not furnished	32000	700	40	11	32750	0.0004188569	1.5998441316
8	Rio de Janeiro	95	2	2	1	11	accept	not furnished	1024	3000	28120	39	32180	0.0002222266	1.4579468587
9	SÃ£o Paulo	80	2	1	1	1	accept	not furnished	875	24000	0	305	25180	0.0003293895	1.260429809
10	Belo Horizonte	260	4	4	6	10	accept	furnished	3200	11900	12500	159	27760	0.0022432505	0.9708131589

f. (5 points) Focus on the five observations with the largest studentized residuals (these are like z-scores so values bigger than 4 or 5 are huge). In two sentences or less, indicate which of the six regression variables seem to be different for those four.

The five observations with the largest studentized residuals have exceptionally high total rental costs, particularly due to unusually large HOA fees and property taxes. Their values for the five regression predictors are not consistently extreme, suggesting that these observations are unusual primarily in the response variable rather than the predictors.

g. (2 points) Refer to part f. Use the Exclude/Unexclude toggle to set aside those five observations and rerun the model. Copy and paste the Summary of Fit table and the Parameter Estimates table in the space below.

RSquare	0.600318
RSquare Adj	0.600074
Root Mean Square Error	2867.13
Mean of Response	5093.893
Observations (or Sum Wgts)	8222

Term	Estimate	Std Error	t Ratio	Prob>|t|
Intercept	307.42588	91.21308	3.37	0.0008*
area	21.195157	0.547639	38.70	<.0001*
rooms	-463.6366	48.03751	-9.65	<.0001*
bathroom	988.70924	47.86946	20.65	<.0001*
parking spaces	641.35093	39.87715	16.08	<.0001*
floor	82.072136	6.24112	13.15	<.0001*

h. (5 points) Refer to part c. In three sentences or less, explain generally what has changed with the fit, and with the numerical value and significance of the parameter estimates.

After removing the five observations with the largest studentized residuals, R^2 increased substantially from 0.0463 to 0.6003, while RMSE decreased from 18,183.54 to 2,867.13, indicating a major improvement in model fit. The parameter estimates changed considerably, particularly for Rooms, Bathrooms, and Parking Spaces, with Parking Spaces changing from a negative to a positive coefficient. All five predictors became statistically significant ($p < 0.0001$), whereas Parking Spaces and Floor were previously nonsignificant.